Stage 2A Sub-Phase Plan: 2A.1 and 2A.2

Development and Benchmarking of an AI-Assisted Optometry-to-OT Translation Approach

This draft proposes splitting Stage 2A technical development (per protocol v0.10, Section 4.3) into two sub-phases, both completed prior to configuration freeze and Stage 2B. Sub-phase 2A.1 establishes an error-informed baseline and selects a candidate model; sub-phase 2A.2 evaluates two complementary refinement pathways—full-context clinical grounding and, if feasible, supervised fine-tuning—before selecting and freezing the final configuration.

2A.1 — Baseline benchmarking

Objective: establish an error-informed baseline prompt—improving on the prompt used to generate the synthetic cases where appropriate—and select a candidate open-weight LLM before investing in full-context clinical grounding or fine-tuning.

1. Retrieve the original synthetic-generation prompt used to produce the Stage 1 dataset.
2. Diff each draft summary against its specialist-corrected reference summary to build an error taxonomy: omissions, hallucinations, terminology drift, risk-interpretation errors, tone/register mismatches.
3. Apply minimal, generic revisions to the baseline prompt to address systematic issues surfaced by the diff, not case-specific fixes.
4. Benchmark candidate locally-deployable/open-weight LLMs against the development set using this revised baseline prompt.
5. Evaluate using multi-LLM-as-judge and majority-agreement (per Section 4.3, to guard against single judge bias)
6. Output: a selected candidate model and a documented baseline prompt, carried forward into 2A.2.

2A.2 — Knowledge-grounded refinement

Objective: reduce the residual gaps identified in 2A.1 by supplying the selected model with the complete case and the complete validated clinical knowledge context within the LLM context window, rather than relying on similarity-based retrieval or prompting alone.

Three validated knowledge layers will be assembled into a compact, structured context and provided together at inference time:

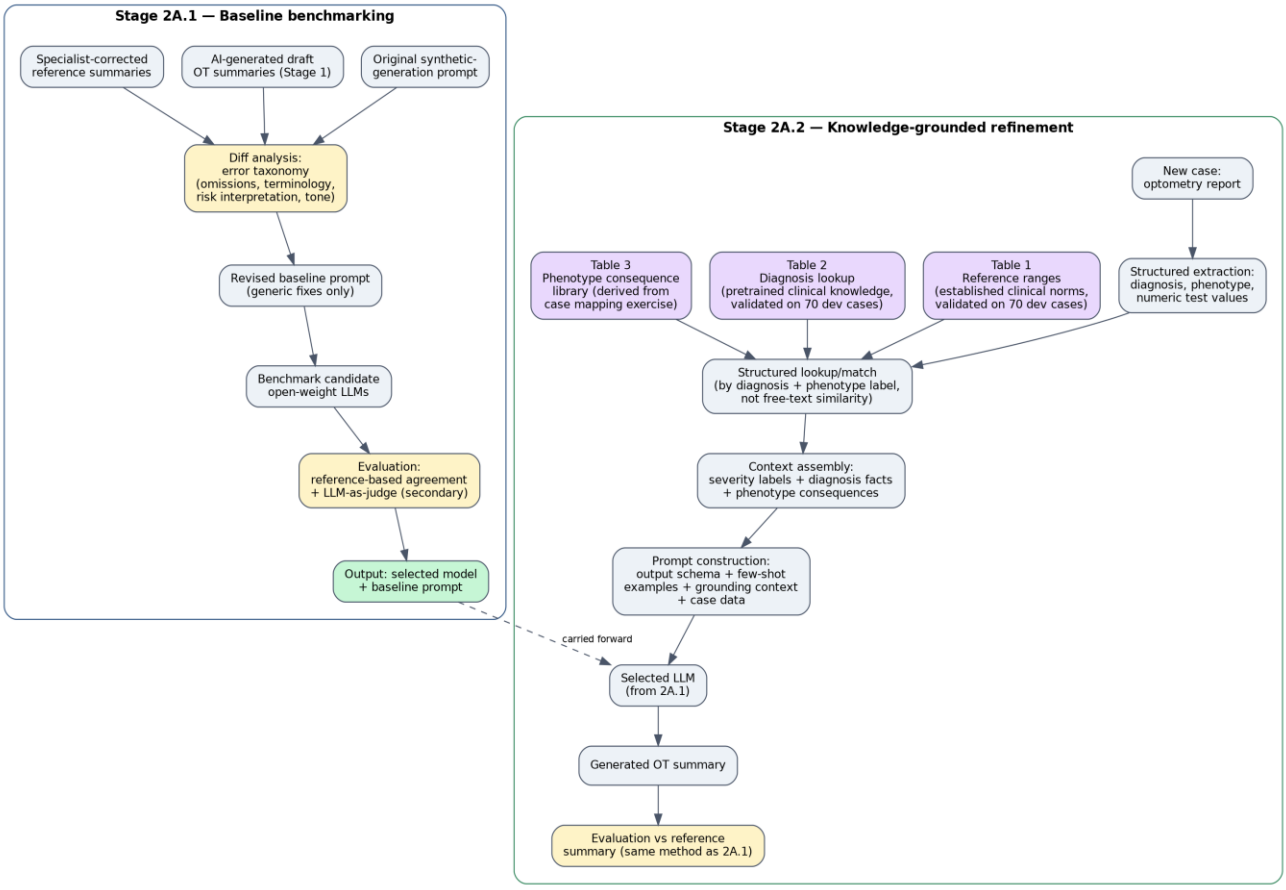
- Table 1: *Reference ranges* which are standard clinical norms for numeric tests (e.g. visual acuity, contrast sensitivity, IOP), giving a normal/mild/moderate/severe band for any measured value. *Sourced from established clinical knowledge* (LLM pre-training data likely contains this), not derived from our cases; the 70 development cases are used only to validate coverage and correctness. This approach ensures our coverage is not limited to the 70 training cases only.
- Table 2: *Diagnosis lookup* that will contain diagnosis label to known direct functional consequence (e.g. sixth nerve palsy -> distance-worse horizontal diplopia). Also sourced from established clinical knowledge and validated, not derived, against development cases (same as Table 1).
- Table 3: *Phenotype consequence* library including broader activities of daily living/participation/environment implications, OTs associate with each primary visual phenotype (not individual diagnoses, to keep coverage generalisable beyond the specific diagnoses seen in development). This layer is derived from the case-mapping exercise itself, since it reflects project-specific clinical judgement with no external reference source.

At inference, the complete source case and all relevant validated entries from Tables 1–3 will be supplied within the model's available context window. We do not propose retrieval-augmented generation (RAG) for this stage. Because the task requires the model to consider the diagnosis, phenotype, numeric findings and their interactions as a whole, retrieval could omit clinically relevant information or introduce context selected for lexical rather than clinical similarity. A full-context design reduces this risk and is feasible only if the combined prompt, case and knowledge tables fit within the selected model's context limit with sufficient output-token allowance. If they do not fit, the preferred mitigation is to compress and normalise the tables without removing clinically material content, then reassess the model or context-window choice.

To-do for Stage 2A.2 (see architecture):

1. Build and clinically validate Tables 1–3 against the 70 development cases, recording coverage gaps, disputed mappings and source provenance.
2. Measure the token budget for the complete input case, Tables 1–3, instructions and expected output; confirm that the selected model can process the full package without truncation.
3. Construct the full-context prompt containing the output schema, instructions, complete validated grounding context and complete case data; use fixed few-shot examples only if token-budget testing shows that they add value without displacing clinically relevant content.
4. Re-run the selected model from 2A.1 with the refined prompt, pipeline, and evaluate using the same method as 2A.1.
5. Compare the full-context configuration with the 2A.1 baseline using the same held-out evaluation set, metrics and multi-judge procedure.
6. Prototype the fine-tuning pathway described below if its minimum data-quality and evaluation requirements can be met within the schedule.
7. Freeze the configuration once performance is acceptable and stable, per Section 4.4, ahead of Stage 2B.

Architecture overview



Proposed for discussion

Finetuning

Time should be reserved in Stage 2A.2 to assess whether parameter-efficient supervised fine-tuning of the selected open-weight model can improve consistency beyond the baseline and full-context approaches. The 70 specialist-corrected cases may support a small feasibility experiment, but they are unlikely to justify unrestricted full-model fine-tuning. The preferred prototype is therefore a parameter-efficient method such as LoRA or QLoRA, subject to the licence, compute environment and governance requirements below.

- Data readiness: confirm that each input–reference pair is complete, consistently formatted, clinically approved and free of unresolved contradictions; define a reproducible transformation into instruction–input–output records.
- Split and leakage control: partition cases by patient/case identity before training, preserve a genuinely unseen test set for final comparison, and ensure no test-case content appears in prompts, lookup tables derived from cases, or training examples.
- Data sufficiency: report diagnosis and phenotype coverage and class imbalance. If the training subset is too small or narrow, treat fine-tuning only as an exploratory feasibility study and do not claim generalisation beyond represented cases.
- Technical requirements: use a licence-compatible open-weight model, secure compute with adequate GPU memory, version-controlled training code, fixed random seeds, checkpointing and complete records of hyperparameters, training duration and cost.
- Safeguards: use conservative learning rates, few epochs, early stopping and repeated runs where practical to limit overfitting; assess hallucination, omission, terminology drift and clinically unsafe interpretation, not only aggregate judge scores.
- Decision rule: compare the fine-tuned model with the unchanged baseline and full-context configuration on the same held-out cases. Adopt fine-tuning only if improvement is clinically meaningful, stable across evaluation methods and does not worsen safety-critical error categories.

RAG

In addition to Knowledge Grounded Refinement, we propose considering a 2nd mechanism: retrieving case-specific few-shot examples (summary) at inference time rather than using look up tables. This is presented separately for discussion, as its viability is less certain and requires prototyping.

- Each development case is indexed as a single linked unit (report + reference summary, keyed by case ID), no chunking, so a retrieved example always carries its correct matching summary.
- Retrieval is two-stage: a hard filter first restricts candidates to the same primary phenotype/diagnosis label as the new case (requires parsing), then a soft similarity ranking (on report text) selects the closest one or two cases within that already clinically-valid pool. This avoids relying on raw text similarity alone, which can rank lexically similar but clinically unrelated cases as “close”.
- The retrieved case/summary pair(s) are injected into the prompt as dynamic few-shot examples.